

Class 09: Candy Mini Project

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Background

Today we are taking a detour to analyze a dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

Data import

The data is all related to halloween candy and is from the 538 website.

```
candy <- read.csv("candy-data.txt", row.names =1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0

Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Nerds") |>
  select(winpercent)
```

```
      winpercent
Nerds    55.35405
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
      winpercent
Kit Kat    76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
      winpercent
Tootsie Roll Snack Bars    49.6535
```

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

```
winpercent
```

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q7. What do you think a zero and one represent for the candy\$chocolate column?

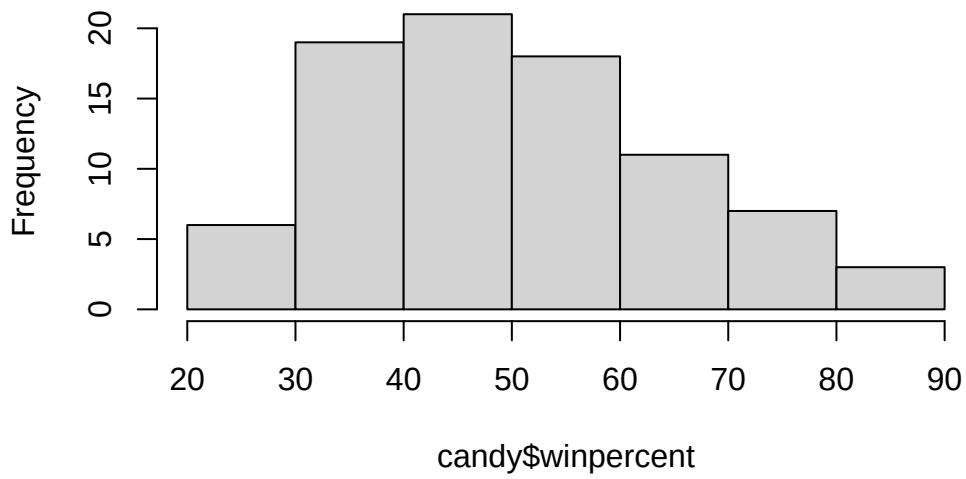
0: not chocolate, 1: yes chocolate

Exploratory analysis

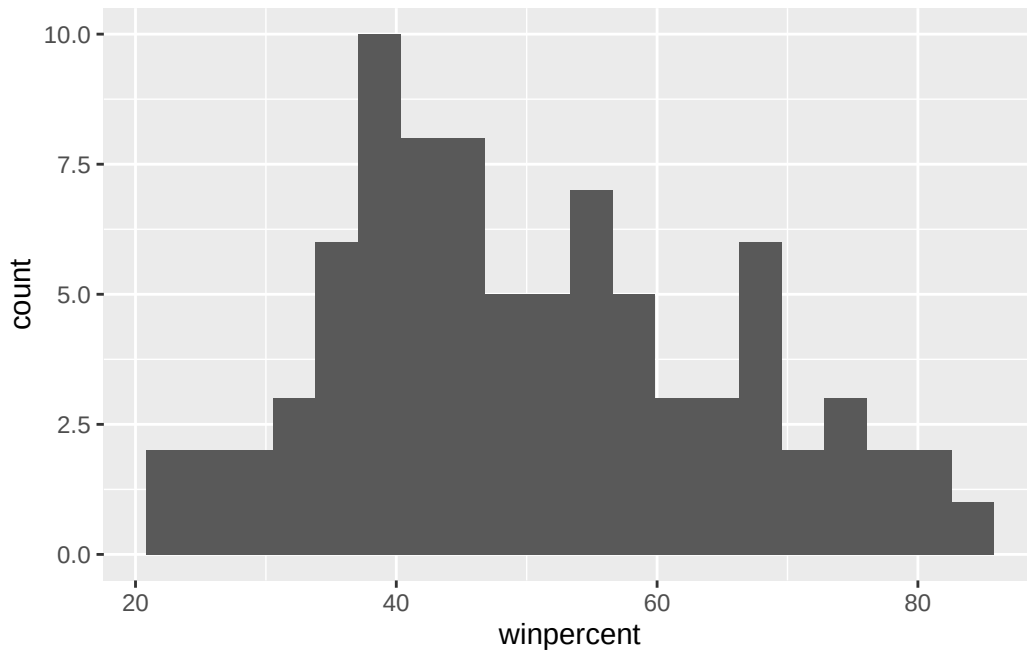
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
library(ggplot2)
hist(candy$winpercent)
```

Histogram of candy\$winpercent



```
ggplot(candy, aes(winpercent)) + geom_histogram(bins=20)
```



Q9. Is the distribution of winpercent values symmetrical?

No

Q10. Is the center of the distribution above or below 50%?

```
median(candy$winpercent)
```

```
[1] 47.82975
```

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc <- candy[as.logical(candy$chocolate),]  
fruit <- candy[as.logical(candy$fruity),]  
mean(choc$winpercent) > mean(fruit$winpercent)
```

```
[1] TRUE
```

Q12. Is this difference statistically significant?

```
t.test(choc$winpercent,fruit$winpercent)
```

Welch Two Sample t-test

```
data:  choc$winpercent and fruit$winpercent  
t = 6.2582, df = 68.882, p-value = 2.871e-08  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 11.44563 22.15795  
sample estimates:  
mean of x mean of y  
 60.92153  44.11974
```

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
candy |> arrange(winpercent) |> head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip		0	0	0		1		0.197		0.976
Boston Baked Beans		0	0	0		1		0.313		0.511
Chiclets		0	0	0		1		0.046		0.325
Super Bubble		0	0	0		0		0.162		0.116
Jawbusters		0	1	0		1		0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
candy |> arrange(desc(winpercent)) |> head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

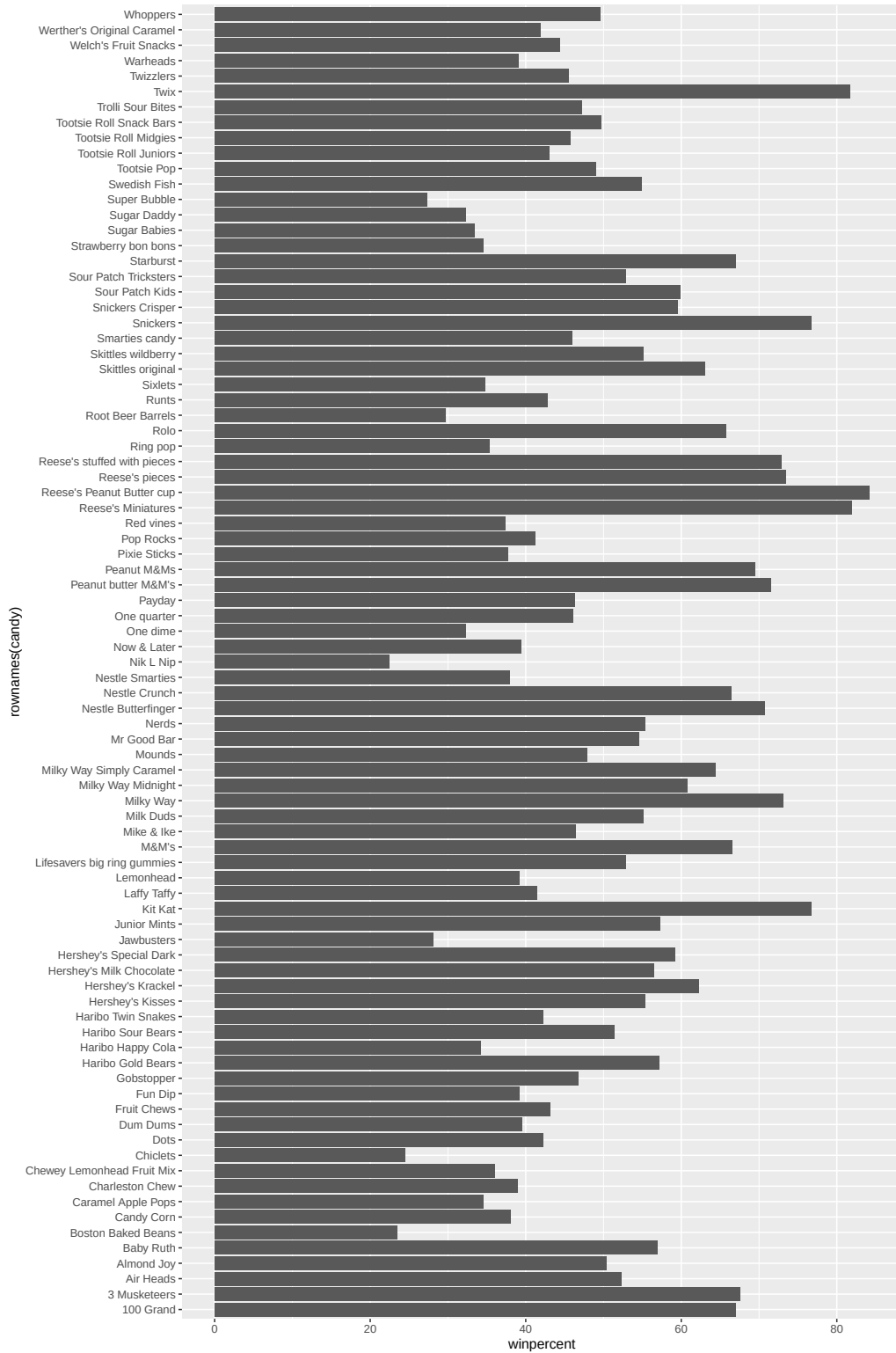
	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup		0	0	0		0		0.720
Reese's Miniatures		0	0	0		0		0.034
Twix		1	0	1		0		0.546
Kit Kat		1	0	1		0		0.313
Snickers		0	0	1		0		0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291

Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

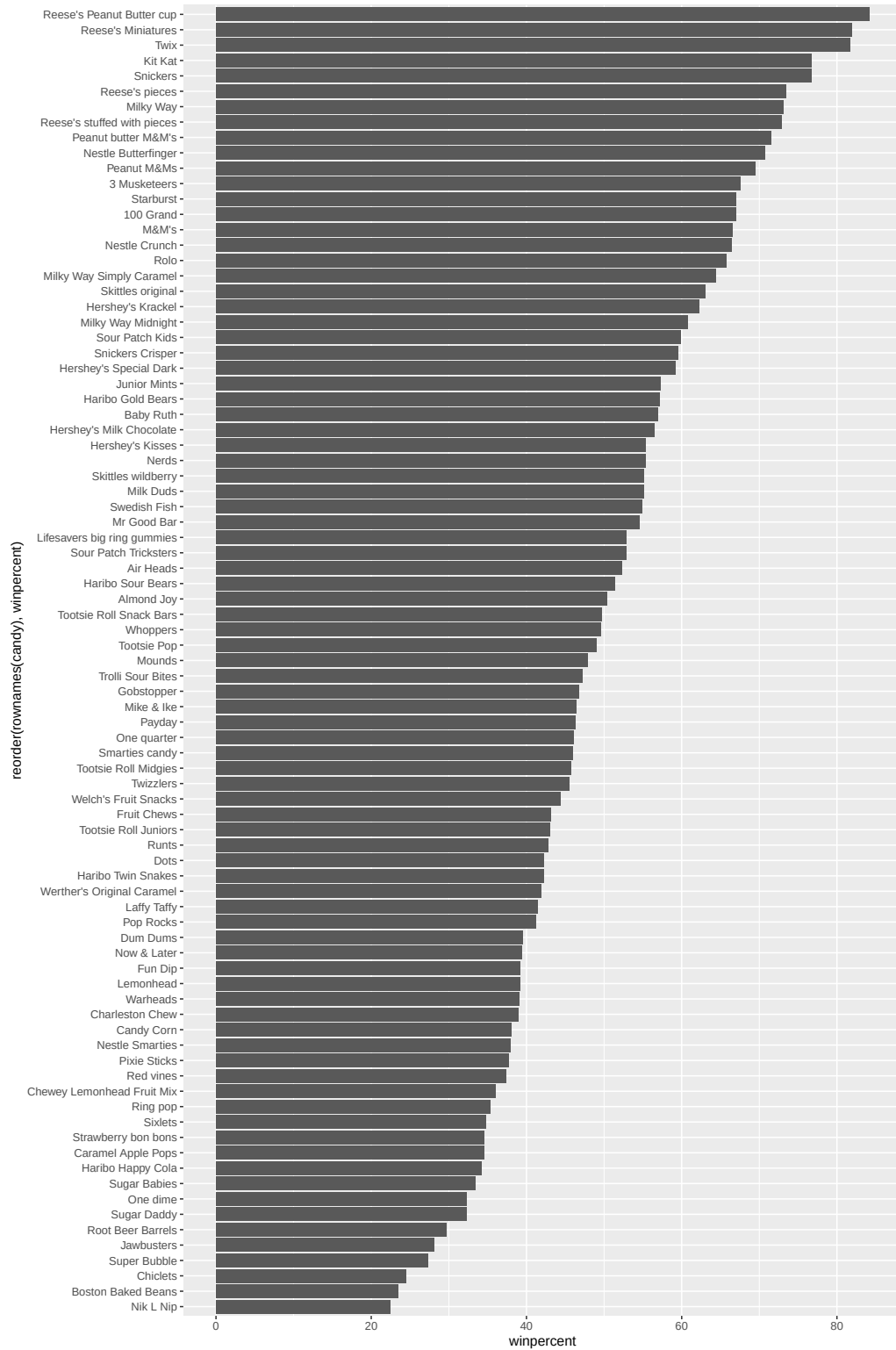
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```

Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy),winpercent)) +  
  geom_col()
```



Adding some color

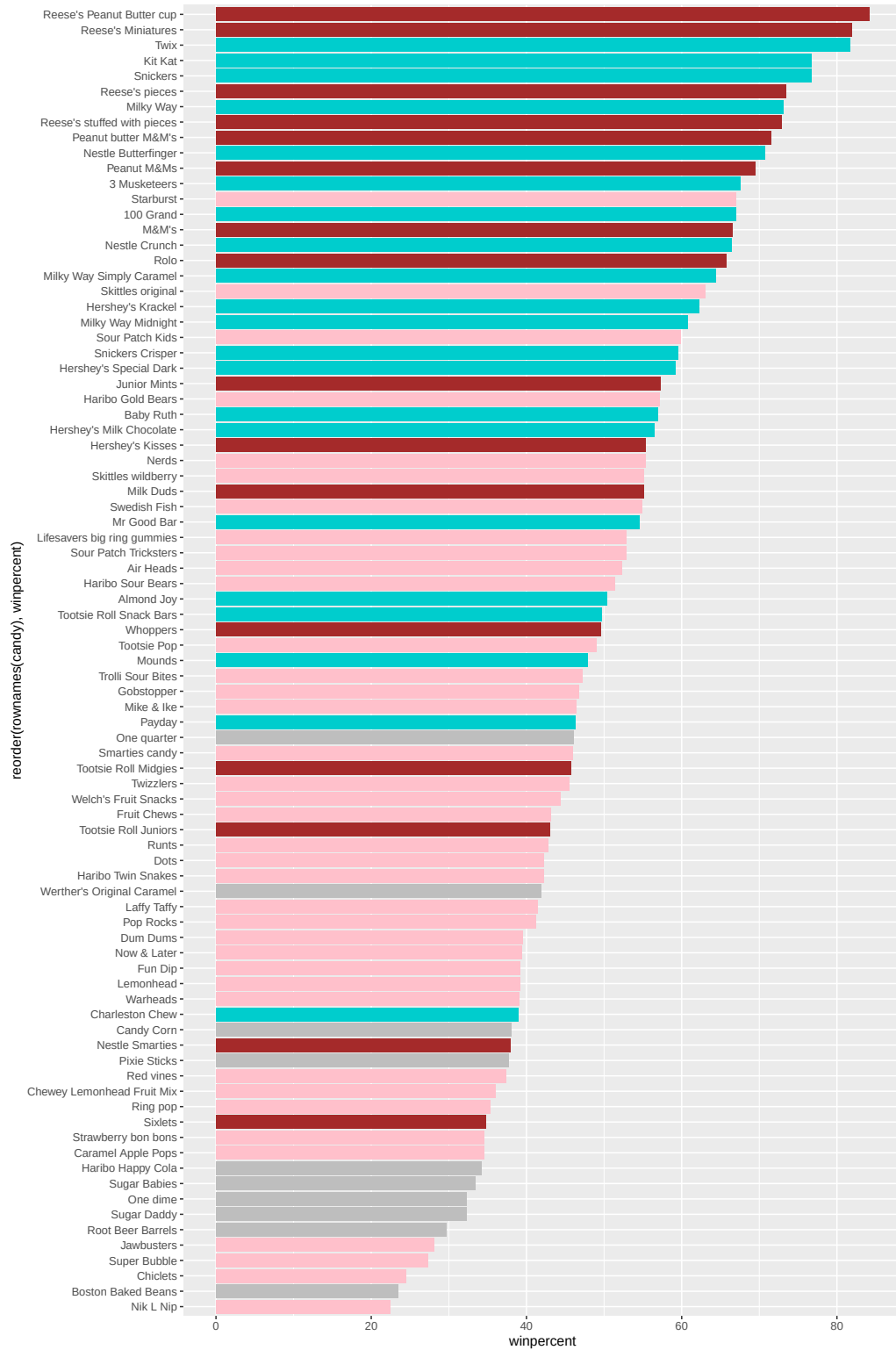
```
mycols <- rep("gray", nrow(candy))

## Chocolate candy
mycols[as.logical(candy$chocolate)] <- "brown"

## Fruity candy
mycols[as.logical(candy$fruity)] <- "pink"

## Bar candy
mycols[as.logical(candy$bar)] <- "cyan3"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill = mycols)
```



Q17. What is the worst ranked chocolate candy?

Sixlets

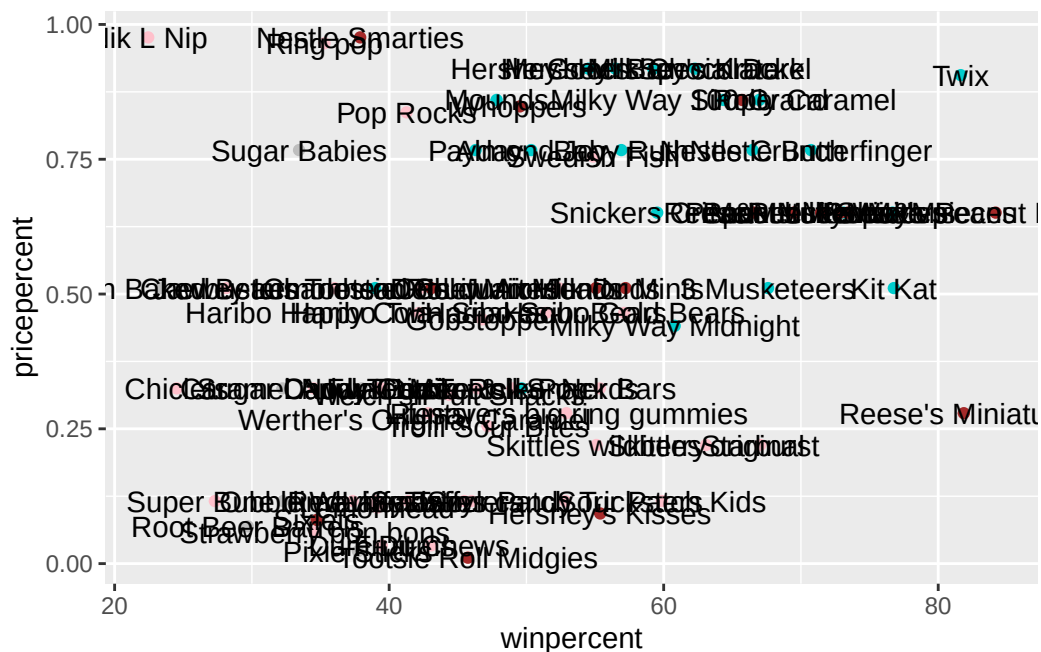
Q18. What is the best ranked fruity candy?

Starburst

Taking a look at pricepercent

Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

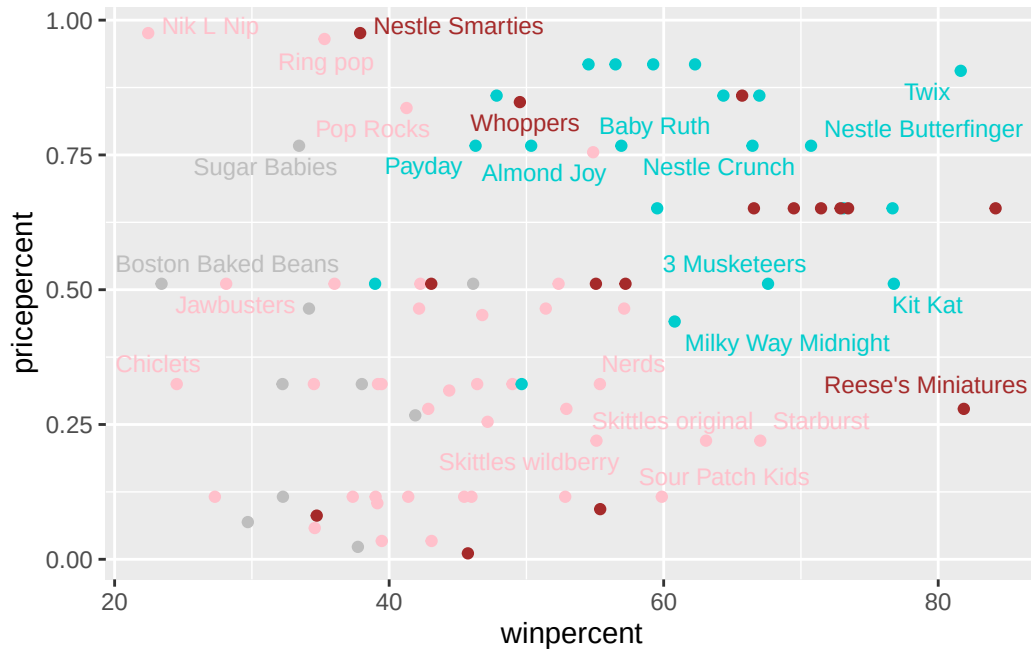
```
ggplot(candy) +  
  aes(winpercent, pricepercent, label=rownames(candy)) +  
  geom_point(col=mycols) +  
  geom_text()
```



We can fix the label text overplotting with an add-on package called **ggrepel** and it's `geom_text_repel()`

```
library(ggrepel)

ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, size=3, max.overlaps = 6)
```



Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

Nik L Nip

```
candy |>
  arrange(desc(pricepercent)) |>
  select(pricepercent, winpercent) |>
  head(n=5)
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Exploring the correlation structure

We can calculate the pair-wise correlation of all our columns

```
cij <- cor(candy)
cij
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
chocolate	1.0000000	-0.74172106	0.24987535	0.37782357	0.25489183
fruity	-0.7417211	1.00000000	-0.33548538	-0.39928014	-0.26936712
caramel	0.2498753	-0.33548538	1.00000000	0.05935614	0.32849280
peanutyalmondy	0.3778236	-0.39928014	0.05935614	1.00000000	0.21311310
nougat	0.2548918	-0.26936712	0.32849280	0.21311310	1.00000000
crispedricewafer	0.3412098	-0.26936712	0.21311310	-0.01764631	-0.08974359
hard	-0.3441769	0.39067750	-0.12235513	-0.20555661	-0.13867505
bar	0.5974211	-0.51506558	0.33396002	0.26041960	0.52297636
pluribus	-0.3396752	0.29972522	-0.26958501	-0.20610932	-0.31033884
sugarpercent	0.1041691	-0.03439296	0.22193335	0.08788927	0.12308135
pricepercent	0.5046754	-0.43096853	0.25432709	0.30915323	0.15319643
winpercent	0.6365167	-0.38093814	0.21341630	0.40619220	0.19937530

	crispedricewafer	hard	bar	pluribus
chocolate	0.34120978	-0.34417691	0.59742114	-0.33967519
fruity	-0.26936712	0.39067750	-0.51506558	0.29972522
caramel	0.21311310	-0.12235513	0.33396002	-0.26958501
peanutyalmondy	-0.01764631	-0.20555661	0.26041960	-0.20610932
nougat	-0.08974359	-0.13867505	0.52297636	-0.31033884
crispedricewafer	1.00000000	-0.13867505	0.42375093	-0.22469338
hard	-0.13867505	1.00000000	-0.26516504	0.01453172
bar	0.42375093	-0.26516504	1.00000000	-0.59340892
pluribus	-0.22469338	0.01453172	-0.59340892	1.00000000
sugarpercent	0.06994969	0.09180975	0.09998516	0.04552282
pricepercent	0.32826539	-0.24436534	0.51840654	-0.22079363
winpercent	0.32467965	-0.31038158	0.42992933	-0.24744787

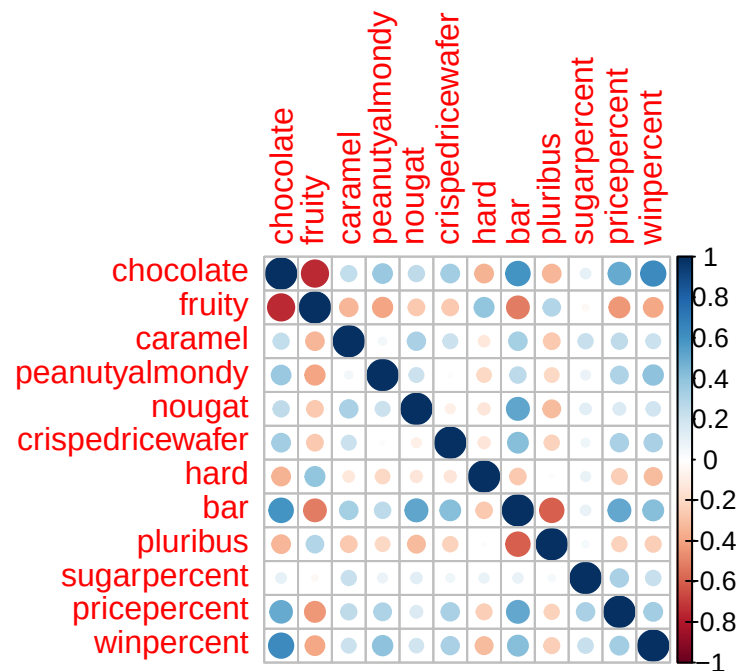
	sugarpercent	pricepercent	winpercent
chocolate	0.10416906	0.5046754	0.6365167
fruity	-0.03439296	-0.4309685	-0.3809381
caramel	0.22193335	0.2543271	0.2134163
peanutyalmondy	0.08788927	0.3091532	0.4061922
nougat	0.12308135	0.1531964	0.1993753
crispedricewafer	0.06994969	0.3282654	0.3246797
hard	0.09180975	-0.2443653	-0.3103816
bar	0.09998516	0.5184065	0.4299293

pluribus	0.04552282	-0.2207936	-0.2474479
sugarpercent	1.00000000	0.3297064	0.2291507
pricepercent	0.32970639	1.0000000	0.3453254
winpercent	0.22915066	0.3453254	1.0000000

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

fruity and chocolate

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

fruity and chocolate

Principal Component Analysis

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

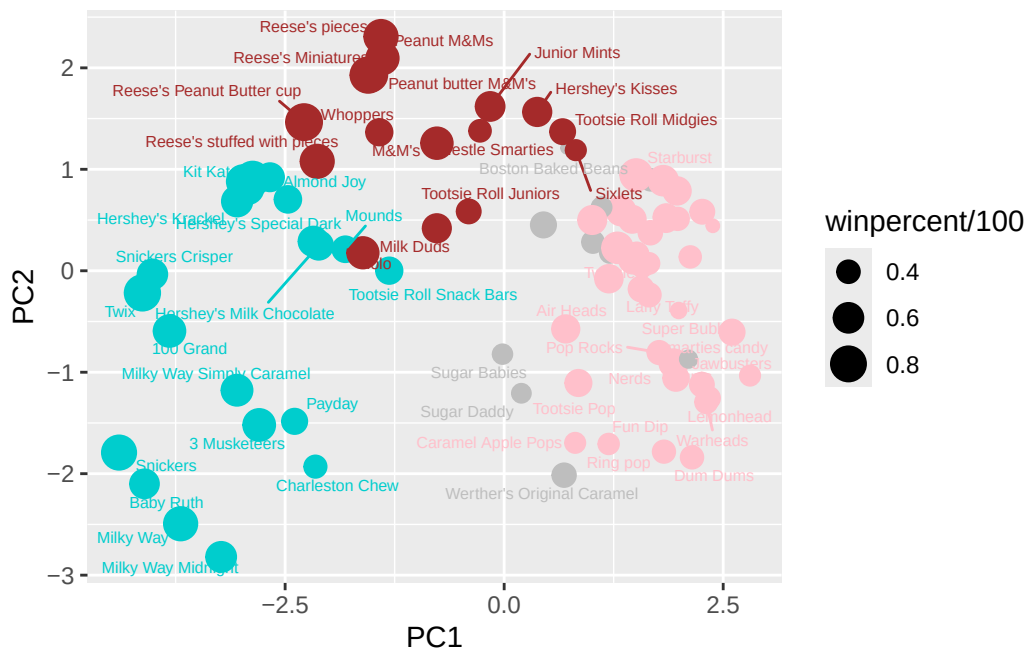
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

First major result figure is the “score plot” of PC1 vs PC2 - how the different candy are related to each other on our new PC axis:

```
my_data <- cbind(candy, pca$x[,1:3])
p <- ggplot(my_data) +
  aes(PC1, PC2, size=winpercent/100, label=row.names(my_data)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, size=2)

p
```



```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

The following object is masked from 'package:stats':

filter

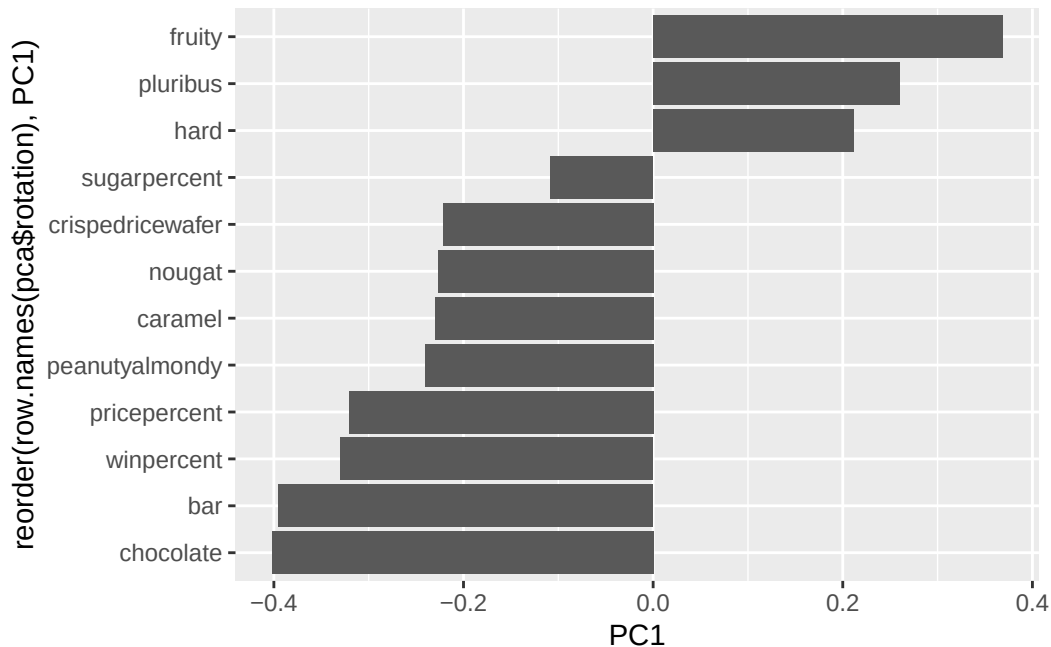
The following object is masked from 'package:graphics':

layout

```
#ggplotly(p)
```

The second major results figure from PCA

```
ggplot(pca$rotation) +
  aes(PC1, reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

fruity, pluribus, and hard are picked up strongly by PC1 in the positive direction. These make sense as this relationship was highlighted by corrrplot previously.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

chocolate candy without fruity flavor (supported by PCA barplot and correlation plot)